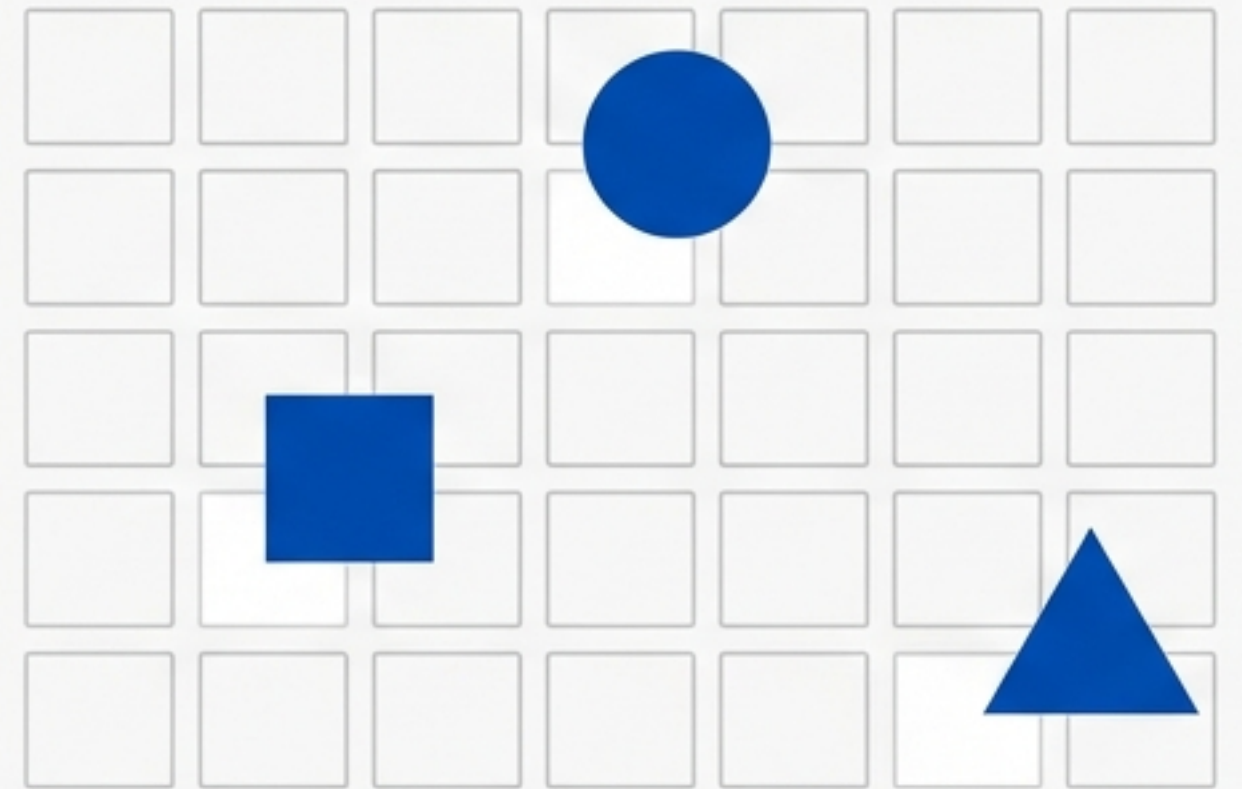


Advanced Missing Data Imputation: Numerical Data

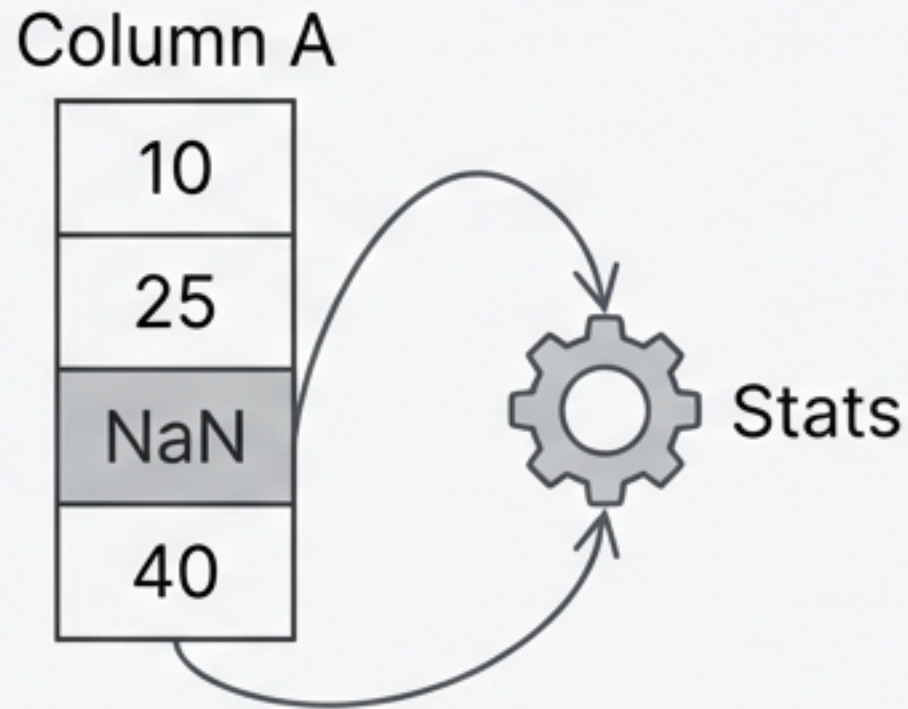
Strategies for Univariate Imputation using Mean, Median, Arbitrary Values, and End of Distribution Techniques.



Defining the Scope: Univariate vs. Multivariate Imputation

Univariate Imputation

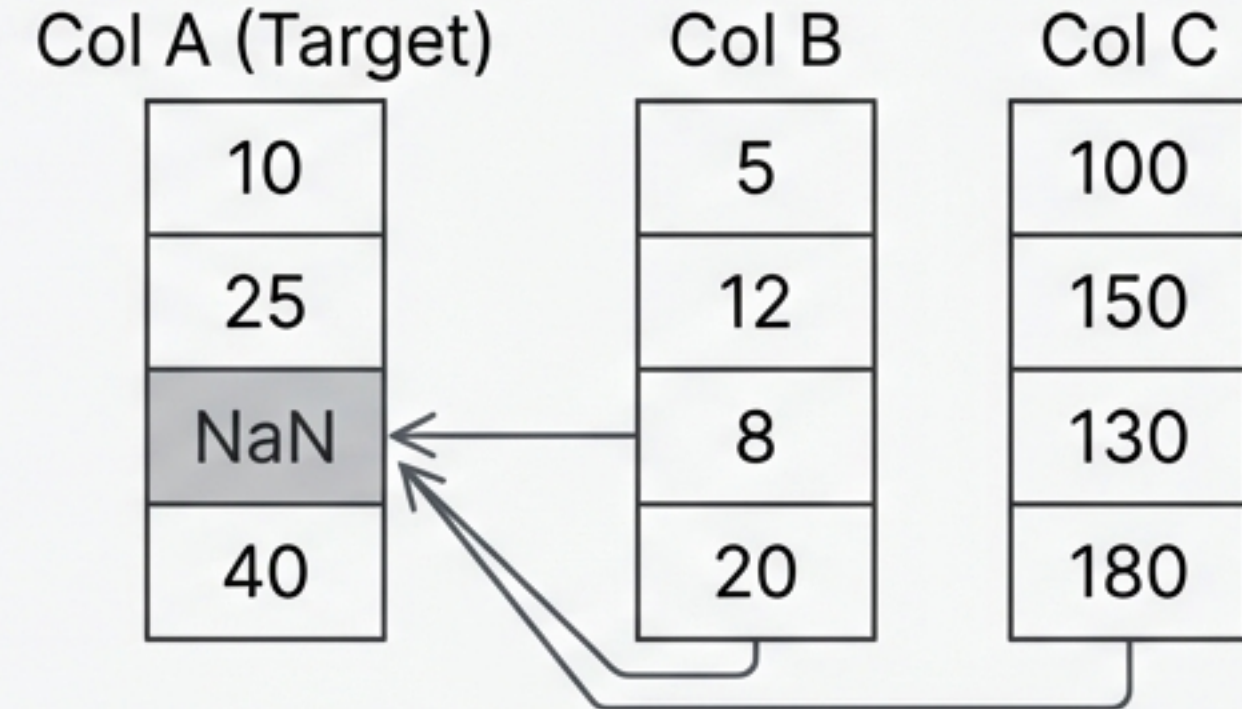
Definition: Filling missing values in a column using only the existing distribution of that same column.



- Mean/Median, Arbitrary Value, End of Distribution.

Multivariate Imputation

Definition: Using correlations with other columns to predict the missing value.



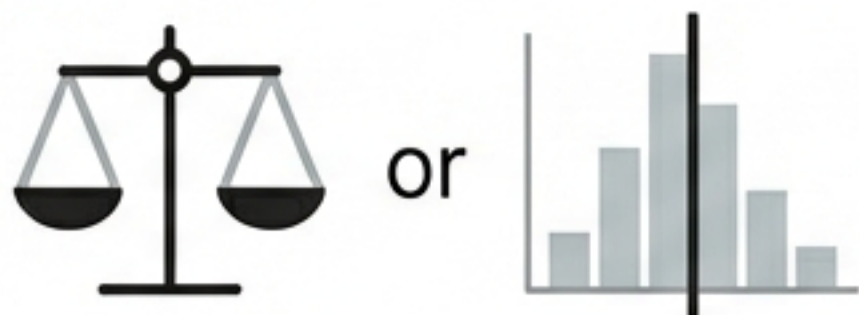
- KNN Imputer, Iterative Imputer.

(Not covered in this module)

This presentation focuses exclusively on Univariate strategies.

The Three Core Strategies

Statistical Average

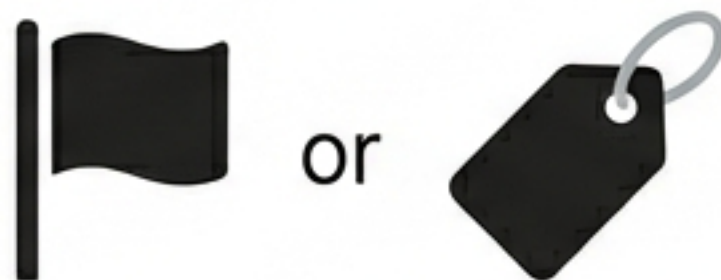


Mean / Median Imputation

Replacing missingness with central tendency.

Best for data Missing Completely At Random (MCAR).

Manual Flagging



Arbitrary Value Imputation

Manually inserting a distinct value (e.g., -1, 99) to differentiate missing data.

Best for data Not Missing At Random (MNAR).

Automatic Flagging



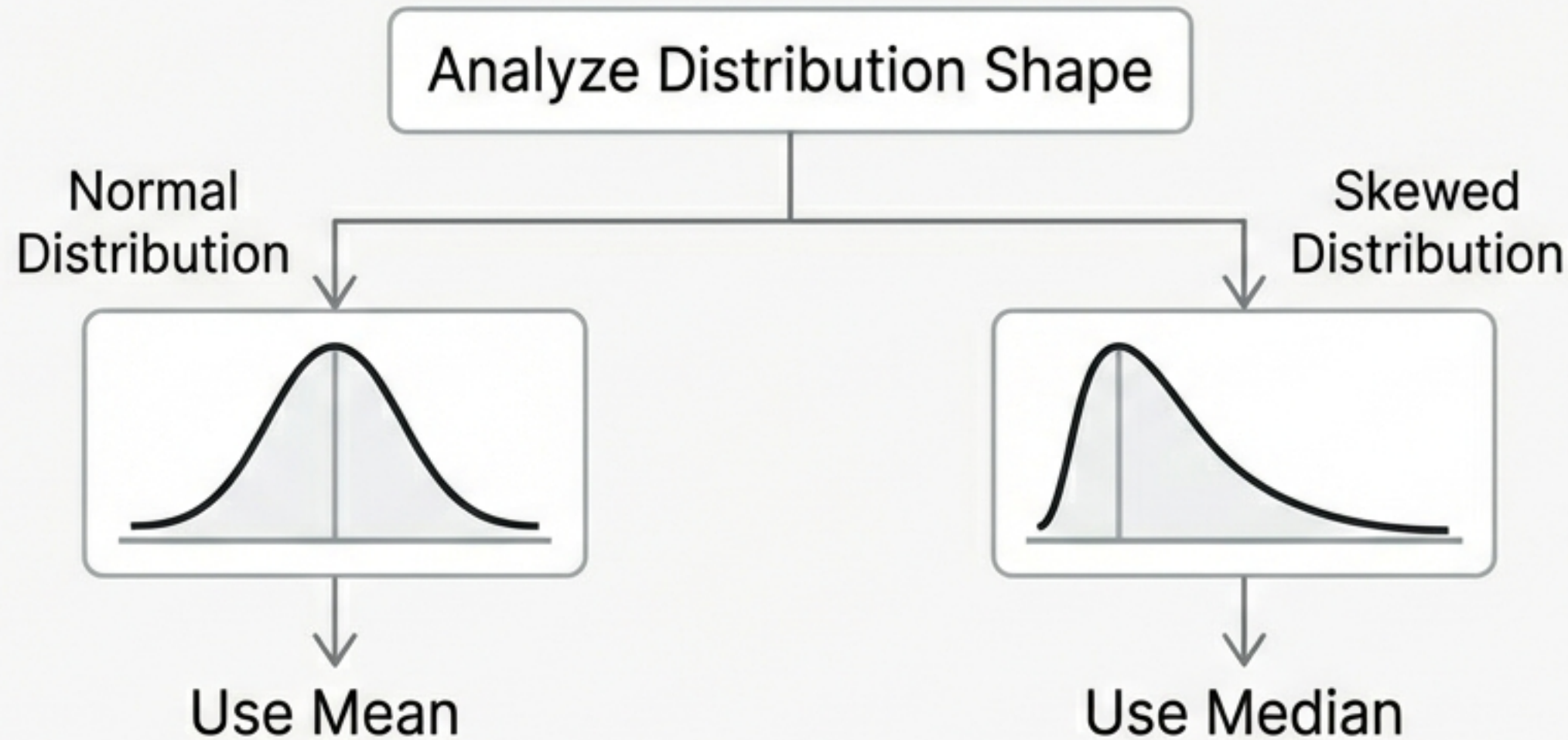
End of Distribution Imputation

Automatically selecting extreme values to flag missing data without manual guessing.

Best for systematic missing data (MNAR).

Technique 1: Mean & Median Imputation

Replacing 'NaN' with the central tendency of available data.



Pros

- Simple to implement.
- Computationally efficient.
- Easy to deploy in production.

Prerequisites

- Data is Missing Completely At Random (MCAR).
- Missing data is $< 5\%$ of total dataset.

The Statistical Cost of Simplicity

How central tendency imputation distorts data integrity.

Distribution Distortion



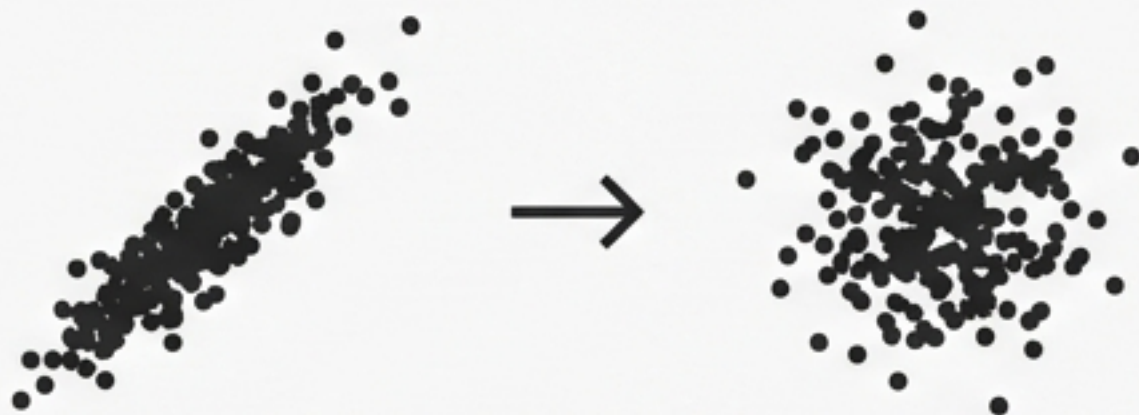
Changes the Shape of the PDF.

Variance Shrinkage



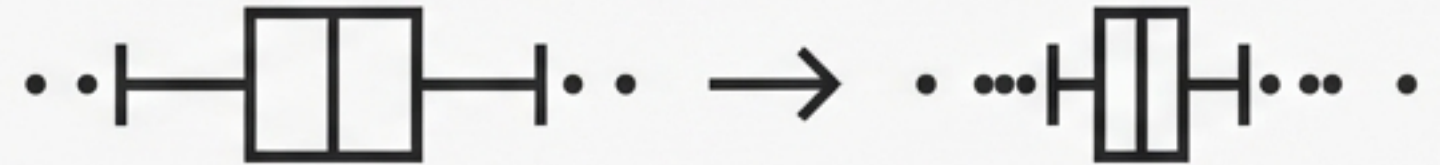
Data clusters tighter; overall variance decreases.

Covariance Shift



Relationships with other variables weaken.

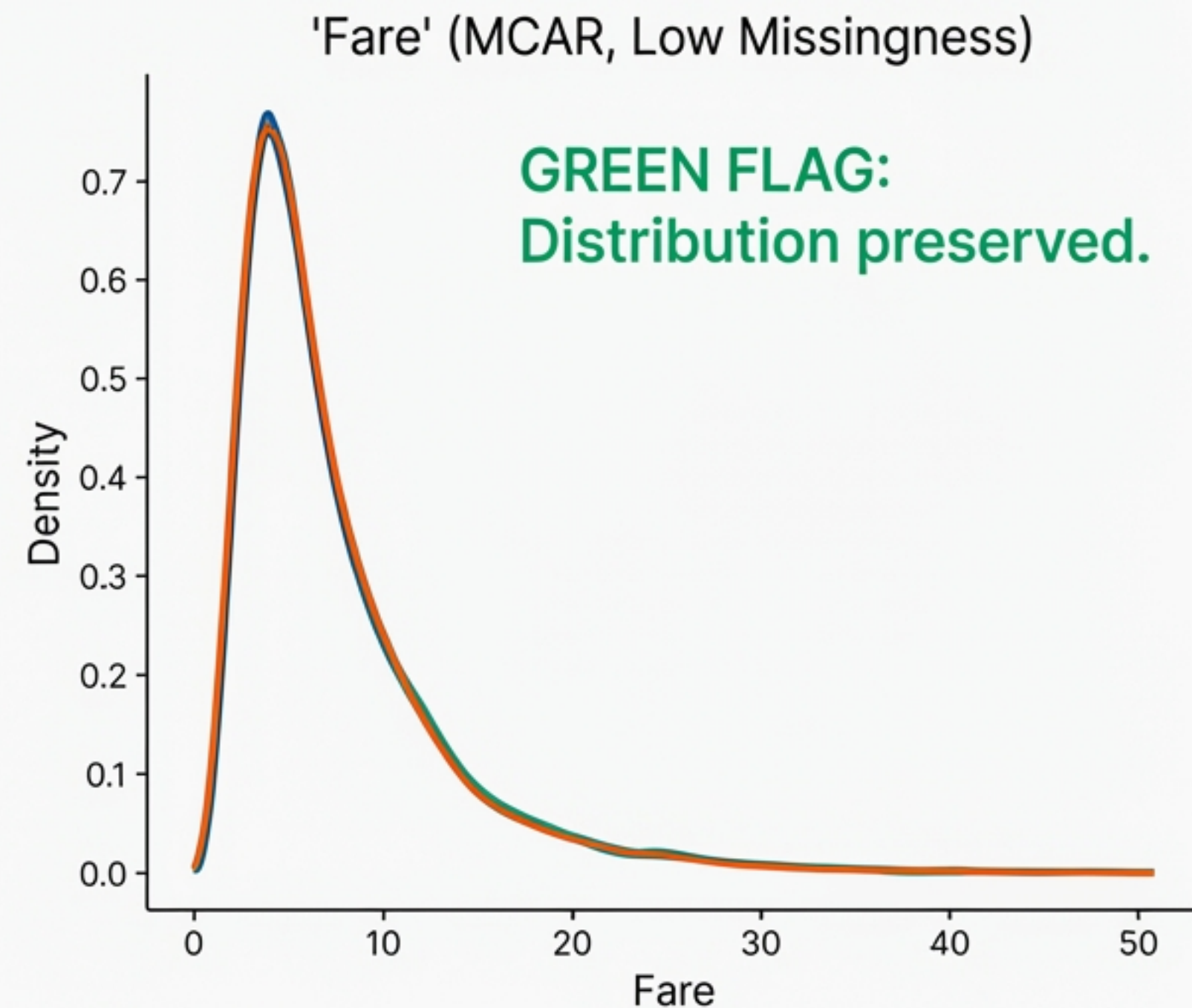
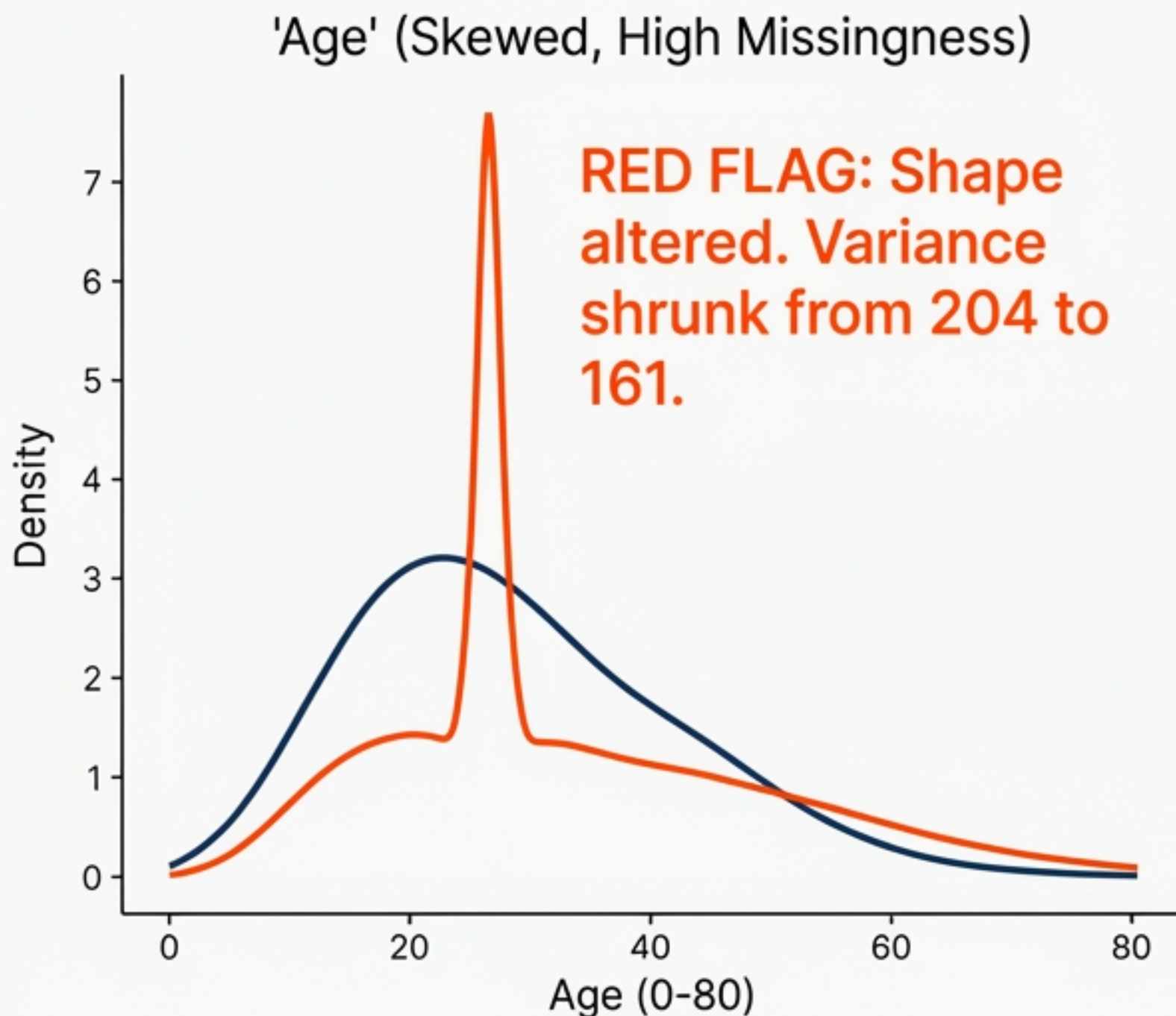
Outlier Creation



Normal values may become outliers in the new range.

Visualising the Distortion: The PDF Overlap Test

Case Study: Titanic Dataset



Insight: If the curves do not overlap significantly, the imputation technique should be rejected.

Technique 2: Arbitrary Value Imputation

Treating 'Missingness' as a feature, not a bug

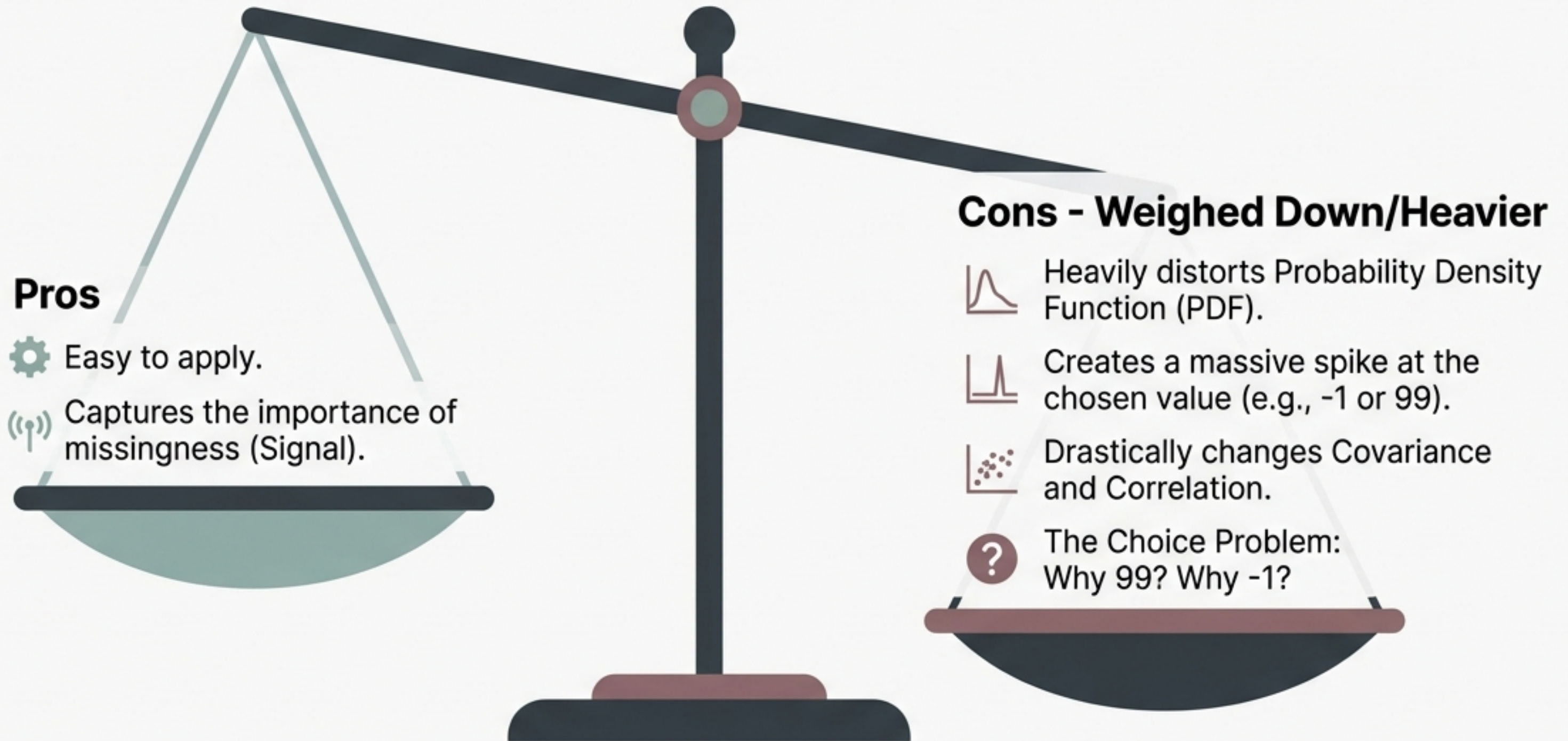
Concept: Replacing NaN with a specific number that does not exist in the distribution (e.g., -1, 0, 99, 999).

Use Case: When data is Not Missing At Random (MNAR). The model needs to explicitly differentiate between "observed" and "missing" records.

Age
22
38
NaN 99
26
NaN 99

Signal: This value is artificial.

The Trade-offs of Arbitrary Values

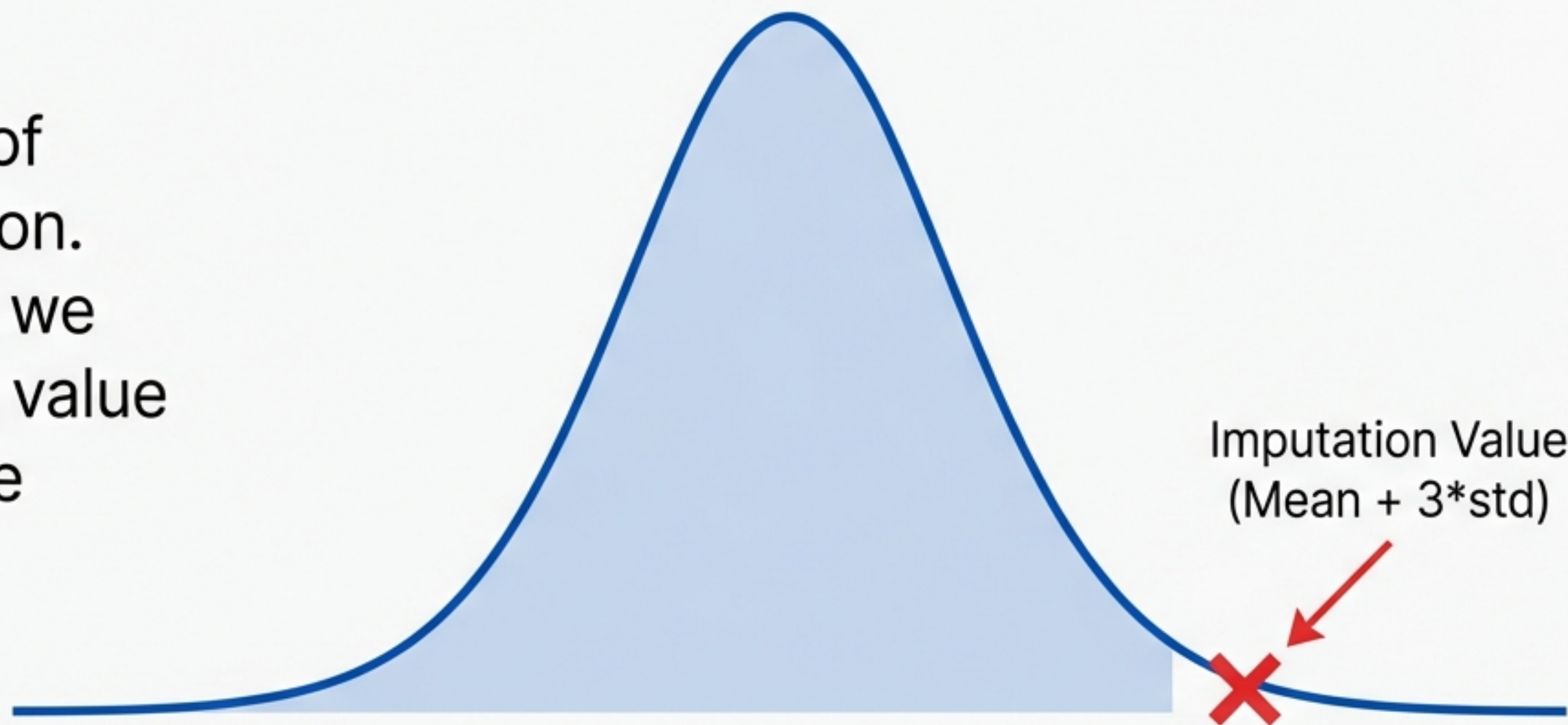


Conclusion: A 'brute force' method. Effective for MNAR but statistically destructive.

Technique 3: End of Distribution Imputation

Automating the selection of arbitrary values.

Concept: An extension of Arbitrary Value imputation. Instead of guessing '-1', we mathematically select a value at the extreme tail of the distribution.



Goal: Flag the value as distinct (outlier) without manual guesswork.

Calculation Rules for End of Distribution

Normal Distribution (The 3-Sigma Rule)

For normally distributed data:

$$\text{Value} = \text{Mean} + 3 \times \text{Standard Deviation}$$

Captures values beyond 99.7% of the population.

Skewed Distribution (The IQR Proximity Rule)

For skewed data:

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR}$$

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR}$$

Uses the Interquartile Range to find statistical extremes.

Implementation: Pandas vs. Scikit-learn

Pandas (`fillna`)

Use Case: Analysis & Quick Fixes

```
df['Age'].fillna(df['Age'].mean())
```

Pros:

Great for visualisation, keeps column names.

Scikit-learn (`SimpleImputer`)

Use Case: Production Pipelines

```
imputer = SimpleImputer(strategy='mean')\nimputer.fit_transform(X_train)
```

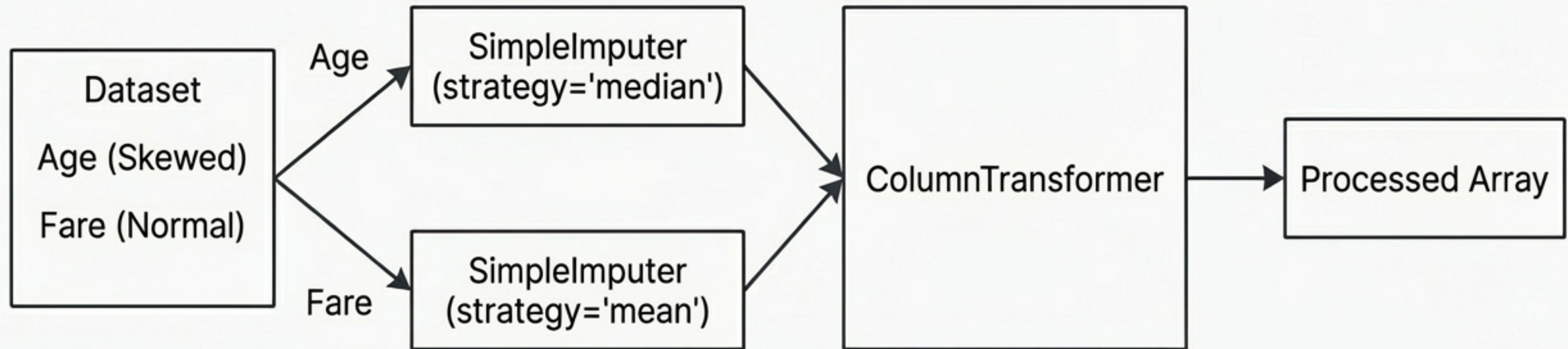
Pros:

Integrates with Pipelines, GridSearch compatible.

Caveat: Returns numpy arrays (loses headers).

Advanced Pattern: The Column Transformer

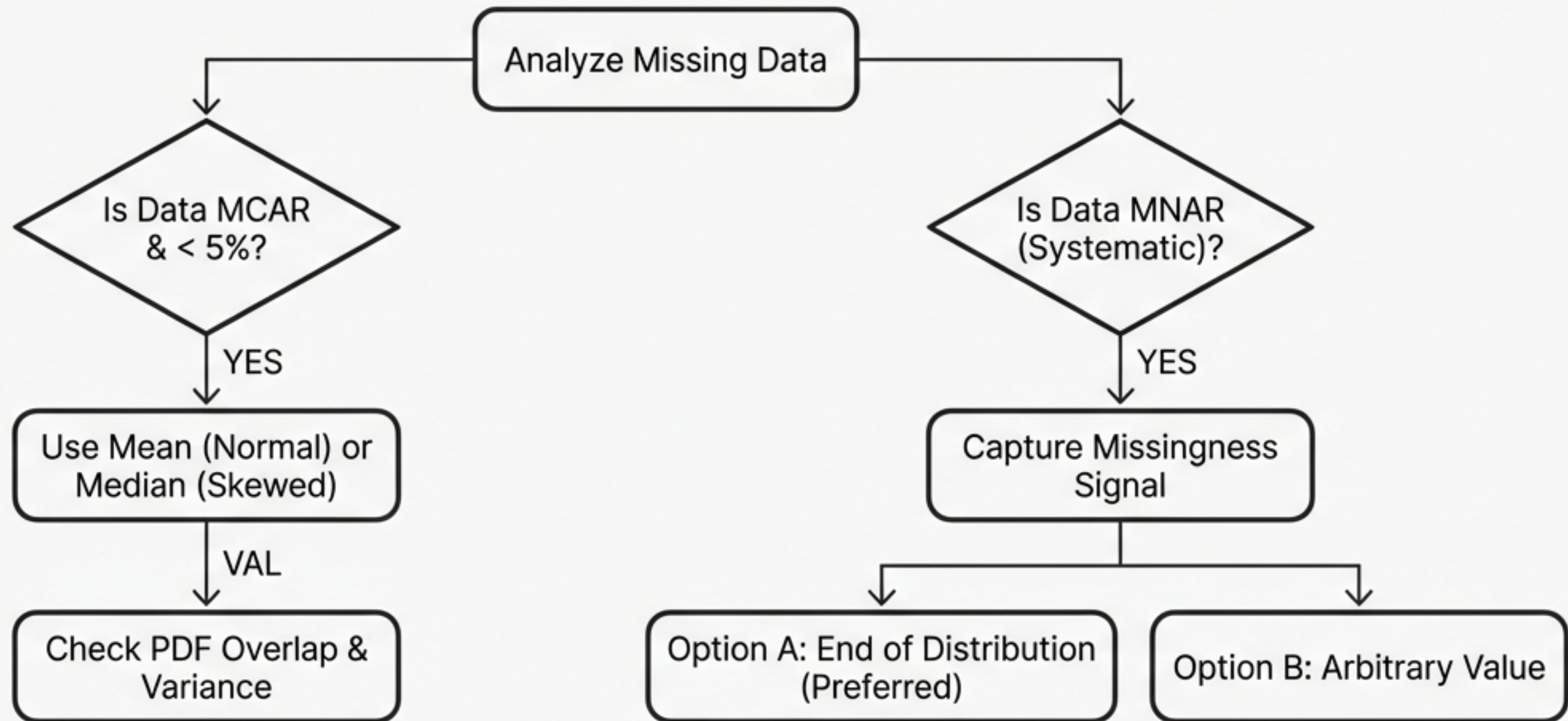
Applying different logic to different columns simultaneously.



Critical Rule: Always fit on Training set, transform on Test set to avoid data leakage.

Strategic Selection Matrix

How to choose the right technique.



Final Check: Monitor Co-variance shifts. If correlations break, consider Multivariate techniques.

Conclusion: The Imputation Trade-off

- ✓ Univariate Imputation is fast but risks distorting data shape.
- ✓ Mean/Median works for random gaps; End of Distribution works for systematic gaps.
- ✓ There is no “One Size Fits All”—visual validation (PDF plots) is mandatory.

Coming Next Module

Inter: Random Sample Imputation (Preserves Distribution Shape)
GridSearch (Automatic Selection)

Thank You

Source Material: CampusX YouTube Channel

Topic: Handling Missing Data | Numerical Data | Simple Imputer